

Essentials of Innovation and Entrepreneurship (EIE) - Unit 1

Overview

Course Foundation

About CIE (Center for Innovation and Entrepreneurship)

- **Established:** December 2017 under Prof. Sathya Prasad
- **Current Space:** Operational since May 2018
- **Mission:** Fueling innovation and empowering students through industry expertise

Key Faculty & Mentors

- **Prof. Sathya Prasad:** Director CIE, Lead Instructor (20 years at Intel, founding President SEMI India)
 - **Raghavendra Deshmukh:** Industry Mentor (SAP, Walmart, Google, Booking.com experience)
 - **Industry Mentors:** 5 experts with 28-35 years experience across cloud, semiconductors, AI/ML, and product management
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Module M0 - Orientation

Learning Framework

Course Objectives: 1. Embrace I&E mindset 2. Kickstart your I&E journey
3. Understand transformative tech-led changes 4. Develop business acumen
5. Prepare for post-university journey

Course Structure

- **Credits:** 2 Credits (2 hours weekly)
 - **Format:** Online with industry expert talks
 - **Assessment:** 2 ISAs (25% each) + 1 ESA (50%)
 - **Grading:** S(90-100), A(80-89), B(70-79), C(60-69), D(50-59), E(40-49), F(<39)
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Module M1 - Introduction to Entrepreneurship

Core Concepts

Entrepreneurship Definition

“The capacity and willingness to develop, organize and manage a business venture along with any of its risks in order to make a profit”

Entrepreneurial vs Entrepreneur

Aspect	Entrepreneurial	Entrepreneur
Role	Works within company	Builds own venture
Ownership	No business ownership	Owns and runs business
Risk	Safety net exists	Full personal/financial risk
Resources	Uses company resources	Builds/raises own resources

Types of Entrepreneurship

1. **SME/MSME** (Small/Medium Enterprises)
 - Local market reach
 - Lower cash flow and risk
 - Gets profitable sooner
2. **IDE** (Innovation-Driven Enterprise)
 - Global market reach
 - Higher cash flow and risk
 - Takes longer to become profitable
 - More clustered in startup hubs

Building Blocks Framework

Innovation = Imagination × Creativity × Entrepreneurship

1. **Imagination:** Envisioning things that don't exist
2. **Creativity:** Applying creativity to generate novel solutions
3. **Innovation:** Applying imagination to address challenges
4. **Entrepreneurship:** Applying innovation, creating value, inspiring others

Startup Ecosystem

Indian Startup Landscape (2024 Highlights)

- Robust ecosystem with government support
- Growing deep-tech sector
- Strong incubator/accelerator network
- Diverse non-tech startup emergence

Startup Definition (India Government)

“An entity working towards innovation, development, and commercialization of new products, processes, or services driven by technology or intellectual property”

Self-Discovery Tools

- **COP Canvas:** Capturing personal Capabilities, Opportunities, and Passions
 - **Ikigai Framework:** Finding your reason for being
 - **Entrepreneurial Traits Assessment:** Self-evaluation tool
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Module M2 - Cultivating an Entrepreneurial Mindset

Mindset Fundamentals

Growth vs Fixed Mindset

Fixed Mindset	Growth Mindset
Abilities are fixed	Abilities can grow
Avoids challenges	Seeks challenges
Gives up easily	Persists through setbacks
Ignores feedback	Learns from feedback
Threatened by others' success	Inspired by others' success

Entrepreneurial Mindset Definition

“A way of thinking that enables you to recognize opportunities, take initiative, and create value despite uncertainty”

Key Entrepreneurial Traits Deep Dive

1. Creativity

- Intersection of projects bringing new ideas and value
- Focus on user improvement and need fulfillment
- Practical application over pure innovation

2. Teamwork

- Coordinated effort toward common goals
- Different skills and perspectives multiplying impact
- Essential for entrepreneurial success

3. Adaptability

- Recognizing when assumptions no longer match reality
- Courage to shift direction without losing momentum
- Embracing uncertainty as constant

4. Dealing with Failure

- Viewing failure as unexpected teacher
- Catalyst for growth and resilience
- Learning from setbacks to improve future performance

Problem-Opportunity Framework

Core Principle: “Don’t Look for Ideas, Look for Problems”

NABC Approach (Stanford Research Institute)

- **Need:** Important client/market need
- **Approach:** Unique solution approach
- **Benefits:** Compelling benefits over competition
- **Competition:** Understanding competitive landscape

“How Might We” Questions

Powerful reframing tool for problem-solving: - “How might we design reusable rockets to reduce space program costs?” - “How might we enable people without credit cards to buy via e-commerce?” - “How might we build foundational Indic LLM model cost-effectively?”

Module M3 - Opportunity Identification & Ethics

Opportunity Identification Framework

Idea vs Opportunity

- **Idea:** Lightbulb moment, flash of inspiration
- **Opportunity:** Open door, pathway to creating value

Opportunity Fundamentals

- Great opportunities often disguised as difficult problems
- Must create significant customer value
- Customer pain = entrepreneur’s value creation source
- Technology alone may not equal business opportunity

Sources of Startup Opportunities

1. **Customer needs** (changing needs)
2. **Personal experience** of problems
3. **Market-driven changes**
4. **Technology-driven changes**
5. **Discontinuities** (social, technological, market)
6. **Serendipity** (“right place, right time”)

Opportunity Evaluation Matrix

Attractiveness Criteria

- **Current Real Need:** Problem exists today
- **Importance:** Problem matters to customers
- **Willingness to Pay:** Customers will pay for solution
- **Solvability:** Can be solved with available/acquirable resources
- **Context:** Industry, market, regulations understood

Value Formula

Opportunity Value =
(Value of Potential) ×
(Number of Target Customers) ×
(Usage Frequency) ×
(Willingness to Pay) ÷
(Current Solution Satisfaction Level)

TIPSC Filter

- **Timely:** Right timing for market
- **Important:** Significant problem
- **Profitable:** Business viability
- **Solvable:** Technical feasibility
- **Context:** Market conditions favorable

Ethics in Entrepreneurship

Business Ethics Foundation

- Ethics = doing the right (moral) thing
- Foundation for integrity and reputation
- Critical for long-term business success
- Especially important for startups (higher vulnerability)

Ethical Leadership Mindset

1. **Integrity:** Honesty, trustworthiness, moral compass
2. **Fairness & Justice:** Impartial treatment of all individuals
3. **Decision Making:** Prioritizing ethical considerations
4. **Responsibility & Accountability:** Taking ownership of actions

Ethics Failure Examples

- **Corporate:** Enron, Volkswagen, Satyam Computers
- **Startups:** Theranos, GoMechanic
- **Consequences:** Scandals, legal trouble, business collapse

Data Ethics Framework

Five key aspects: 1. **Ownership:** Clear data ownership rights 2. **Transparency:** Open about data practices 3. **Privacy:** Protecting personal information 4. **Intention:** Ethical purpose for data use 5. **Outcomes:** Responsible impact management

Practical Applications

Post-Session Activities

1. **Failure Resume:** Document and learn from past failures
 2. **Problem Identification:** Using UN SDG goals as inspiration
 3. **Napkin Pitch:** Quick opportunity validation framework
 4. **NABC Analysis:** Structured opportunity evaluation
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Integration with PES University Ecosystem

Innovation Journey Timeline

- **Semester 3-4:** EIE-I & EIE-II (Learn & Practice)
- **Summer Programs:** CIE/CoE Internships
- **Semester 5-6:** Capstone Projects (Prototype Development)
- **Semester 7-8:** Incubation & Acceleration

Support Programs

- **CIE Spark 2025:** Problem validation and team formation
 - **CIE Ignite:** Advanced incubation program
 - **Startup Nation Olympiad:** Global competition platform
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Key Learning Outcomes

By completing this unit, students will:

1. **Develop Entrepreneurial Mindset:** Understand and cultivate growth mindset, creativity, adaptability, and resilience
2. **Master Opportunity Identification:** Use systematic approaches like NABC and TIPSC to evaluate business opportunities
3. **Apply Ethical Framework:** Integrate business and data ethics into entrepreneurial decision-making
4. **Build Practical Skills:** Complete hands-on activities for self-assessment,

problem identification, and opportunity validation

5. **Prepare for Innovation Journey:** Understand the pathway from idea to market and available university resources

This unit establishes a strong foundation for entrepreneurial thinking while emphasizing the critical importance of ethical business practices in today's startup ecosystem.